Fluid Flow Through Fluidized Bed

Experiment 6
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Executive Summary

Fluidized beds are mainly used for reaction purposes. The reaction engineering of some processes will require a fluidized bed reactor, for the reaction between the raw materials to reach completion. The use of a fluidized bed reactor allows for the interaction between a solid catalyst and gas.

The objective of the Fluid Flow Through Fluidized Beds is to measure the gas pressure drop across a bed of particles, and to observe the fluidization behavior of two different types of particles. The two different particles are fine silica sand and ion exchange resin beads. The particles are packed into the bed and then the air is turned on, allowing the gas to flow through the bed, and therefore effecting the pressure.

The results of this experiment were as expected. The pressure drop was measured at both increasing and decreasing pressures to obtain the set of data. A total of 4 trials were completed, 2 for the silica sand and 2 for the resin beads. The trials for each particle consisted of a 6 inch and 10 inch packed bed. The minimum flowrate of fluidization for the sand in the 6 inch bed is approximately 60.725 L/min. The minimum flowrate of fluidization for the sand in the 10 inch bed is larger, as expected, at approximately 64.33 L/min. Similarly, the minimum flowrate of fluidization for the resin beads in the 6 inch bed is approximately 75.161 L/min. The minimum flowrate of fluidization for the resin beads in the 10 inch bed is approximately 78.77 L/min. The trends between increasing and decreasing the pressure in each trial are quite similar. The values mentioned above are reasonable as the particle size of the resin beads are larger than the silica sand, 841 microns and 500 microns respectively, and the size of the particles play a role in the flowrate.

Overall, the *Fluid Flow Through Fluidized Beds* lab was successful. The results show similar trends when increasing and decreasing the pressure in each of the 4 trials and that the flowrates for the particles in the 10 inch beds were larger than the that of the 6 inch beds.

Introduction

The purpose of this experiment is to measure the gas pressure drop across a bed of particles. The particles that are used are silica sand and ion exchange resin beads, which will need to be packed into the fluidized bed vessel. There are four trials that this experiment requires.
Each type of particle will have two trials, one completed when the particles are packed to a height of 6 inches, and the other when the height is 10 inches. The multiple trials will allow for the effects of the particle size, height of the bed, and the gas velocity on the bed of particles in the fluidized bed vessel to be observed.

Fluidized beds are a large part of reaction engineering. The solid particles in the vessel can become fluid-like when a gas flowing through in an upward direction creates a drag force that's greater than the force of gravity [4]. The advantages of using a fluidized bed reactor include exceptional heat transfer, and the ability to move solids as if they were a fluid [4]. The disadvantages of using a fluidized bed reactor is that there is a greater possibility of erosion than in other types of reactors, and the bubbles produced in the fluidized bed need to be controlled, as larger bubbles can reduce the mass transfer between the phases [4]. This experiment presents the basics on how fluidized bed reactors operate.

For this lab, the sand or resin particles are packed into the fluidized bed vessel to a height of 6 inches. The air is then turned on and begins to flow through the bed. The pressure drop is recorded in increments as the gas velocity in the vessel is increased until steady fluidization occurs. The same procedure is followed when the pressure is decreased back to zero. The process is then repeated at 10 inches for the sand, and then again at both 6 and 10 inches with the resin beads. The void fraction for each particle also has to be determined. This can be achieved by mixing a known amount of the particles with a known amount of water. The water displacement level is then recorded and the void fraction can be calculated.

Theory

Fluidized Bed:

In fluidized bed experiments, there are a multitude of particle types used for column packing. They can be as simple as sand, or more complex particles such as those with undefined geometry [1] and even catalysts for various reaction applications. Particle sizes can also vary in size, although they usually fall within the range of 50-500 μm.

The behavior of the particles is absolutely dependent on the carrier gas velocity as evidently presented in Figure 1.

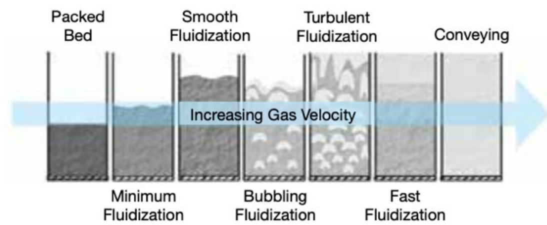

Figure 1: Fluidization shift with increasing gas velocity [2]

From Figure 1, it shows that as the gas velocity slowly increases, the stationary packed bed will remain fixed until there is enough velocity carried by the gas to lift the weight of the particles in the bed. This is known as the minimum fluidization (or superficial) velocity (V_o) [3]. When this happens, a pressure drop is also achieved due to the governing Ergun equation which describes the pressure drop relationship:

$$\frac{\Delta P}{L} = \frac{150 V_o \mu}{\Phi_s^2 D_p^2} \frac{(1-\varepsilon_m)^2}{\varepsilon_m^3} + \frac{1.75 \rho V_o^2}{\Phi_s D_p} \frac{(1-\varepsilon_m)}{\varepsilon_m}$$

Where:
ΔP : Pressure Drop across packed bed (Pa)
L : Bed Height (m)
ρ : Fluid Density (kg/m^3)
ε_m : Bed Voidage at Minimum Fluidization
Φ_s : Sphericity of Particle
μ : Fluid Viscosity (Pa.s)
D_p : Particle Diameter (mm)
V_o : Superficial Velocity (m/s)

Additionally, the superficial velocity, V_o, can be calculated by:

$$V_o \approx \frac{g(\rho_p - \rho)}{150\mu} \frac{\varepsilon_m^3}{1-\varepsilon_m} \Phi_s^2 D_p^2$$

Where ρ_p is the particle density.

For the purpose of this experiment, it was assumed that the sphericity of the particles was uniform ($\Phi_s = 1$), and that the sand and resin beads used were sufficiently small such that the particle's Reynolds Number ($Re_p < 1$). It was also assumed that the velocity drag force and the bed gravity were equal to each other. As a result, the equations listed above were made suitable for the purpose of calculations.

As the velocity of the carrier gas increases, it will eventually come to a point where it achieves the highest pressure drop possible. This is known as the minimum fluidization point (or incipient fluidized bed), and if the flow rate of the gas continues to be increased, it will then go onto the smooth, bubbling, turbulent, and fast fluidization stages as depicted in Figure 1. Beyond the minimum fluidization point, no observable pressure drop can be attained from there on and the gas flow simply passes through the packed bed in the form of bubbles.

It is also important to note the relationship between the pressure drop across the fluidized bed and the superficial velocity of the carrier gas. In Figure 2, the upward and downward pointing arrows represent the increasing and decreasing gas velocities respectively. Point B is where maximum pressure drop occurs, and it plateaus past the minimum fluidization velocity.

Additionally, it is also worth mentioning that the pressure drop for increasing and decreasing gas velocities differ to an extent. This hysteresis is caused by the fact that the system needs to overcome the friction coefficient for the static bed to activate the particles into motion with increasing gas flow; and it is not observed with decreasing gas flow.

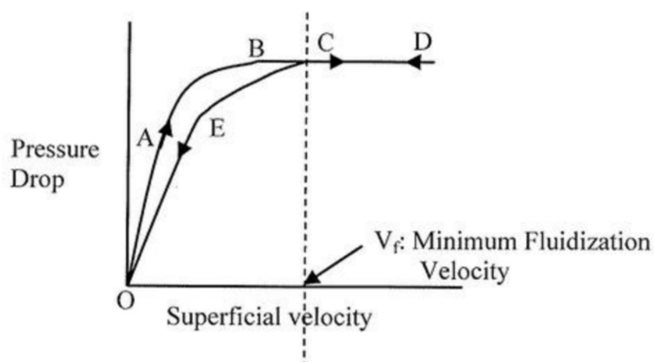

Figure 2: Relationship between Pressure Drop and Superficial Velocity

Particles:

Two properties of particles were needed for calculations later on. They are:

1. Average Particle Diameter

   To find the average particle diameter ($\overline{D_p}$), the particles were sifted through various sieves until they showed minimal resistance in extrication. That is, the point where it there was just enough pressure applied to sieve the particles through the sifter. The following equation was then employed to determine $\overline{D_p}$:

$$\overline{D_p} = \frac{1}{\sum\left(\frac{x_i}{D_{pi}}\right)}$$

   Where:
   x_i : mass fraction
   D_{pi}: Particle Size based off sifter used (mm).

2. Void Fraction (ε_m)

   To 100 mL of sand in a graduated cylinder was added 100 mL of water. The suspension was allowed to sit for a few minutes, and the total amount in the cylinder was recorded after. The values obtained from the readings were then substituted in the equation below:

$$\varepsilon_m = \frac{\text{Total volume in cylinder - volume of water added}}{\text{Total volume of water and particle}}$$

   This is a dimensionless variable since both the numerator and denominator carried the same units. The void fraction of sand was found to be (.375) and (.444) for the resin beads.

Results

The goal of this experiment was to simulate a fluidized bed utilizing both sand and resin beads. Throughout the experiment the pressure being run through the bottom of the system was varied. As the pressure of the stream was increased the particles started to move around and the bed height was shown to increase. Once the pressure reaches a large enough point the particles will begin to move around freely and will simulate a liquid. This is due to stiff bubbles that play a role in mixing the solids which make them look liquid like.

Pressure drop and its effects are often measured as a flow rate. In order to observe the effect from pressure drop, the Ergun equation can be used. It is important to keep in mind that the Ergun equation can only be used at low pressures.

As shown in figure 3, the minimum pressure to achieve fluidization is seen to be at approximately 60.725 L/min. At this point the bed is completely fluidized. The minimum flow rate of gas is determined when the pressure drop of the gas became steady and no longer increased or decreased. The pressure drop was read from a manometer that had an error of around 0.1 inH_2O. This is due to human error and the manometer fluctuating after the fluidization point. It is also clear that in figure 3, the curves are not the same when increasing and decreasing flow rate. This is due to there being friction within the system when first setting the particles into motion. This variable will not need to be overcome when starting at a high flow rate and lowering to zero.

Knowing the minimum flow rate needed to fluidize the bed, the minimum fluidization velocity can be calculated. This was done by dividing the flow rate of air and dividing it by the cross section area of the fluidized bed. The cross section was found to be around .008 m^3.

When comparing the minimum fluidization velocity for the 6 inch and 10 inch bed of sand, it is observed that the gas flow rate needed to achieve fluidization is higher with a larger bed. The fluidization point for the 10 inch bed of sand, is at approximately 64.33 L/min. This agrees with what was predicted due to there being more mass of sand. This will require a larger “force” in order to set the object in motion. This force is being applied by the flow rate on the bottom of the particle bed.

When comparing the minimum fluidization velocity of the sand to the resin beads there is a clear difference when comparing the stream velocities at fluidization point. Comparing the 6 inch beds, the sand had a fluidization velocity of 60.725 L/min while the resin beads had a minimum fluid velocity of 75.161 L/min. This is due to the size of the particle. A larger particle will take more air to move around and to fluidize the bed. Therefore, the flow rate of air applied to the bed would need to be larger in order to get the larger resin beads to their fluidization point.

It is also important to recognize that the bed height increases as the flow rate of air is increased. This is due to the stiff bubbles that run through the beads or the sand that create more movement which then leads to fluidization. A larger flowrate of air leads to more stiff bubbles which then leads to more movement which in turn increases the bed height. This can be shown in figure 7.

Discussion

The Fluidized Bed experiment consisted of finding the pressure drop over a packed bed column while changing air flow rate to the column. Fluidization can be physically seen by when the packed material appears to behave like a moving liquid. Air was provided evenly to all parts of the packed bed, so all trends can be assumed to constant across a cross section of the column.

Figure 3 shows the trends for a column packed 6 inches high with a fine sand. As the flowrate of compressed air was increased, the pressure drop was also observed to increase at almost a totally linear rate. The initial pressure drop for this case was measured to be slightly above zero; however, immediately after air flow was introduced the pressure drop then started to rise at a linear rate. Since when fluidization occurs the pressure drop stays constant, even with increasing flow rates, the overall trend plateaued here. While partial fluidization was observed early on, the data perspective shows it starting later when the packed material seemed to be completely liquid. A small decrease in pressure drop was recorded in the middle of raising the flow rate of air. This occurred because when the flowrate system was being adjusted, the packed bed material was allowed time to settle and the pressure drop lowered. Once fluidization was reached, lowering the air flowrate showed more smooth decrease in pressure drop. In the case of the 6-inch packed sand, the pressure drop did not decrease all the way to zero, most likely due to the packing of the sand at the end of the trial.

Figures 4, 5, and 6 show a similar trend as there was for the 6 inches packed sand. For all cases, 6/10-inch packed sand and resin, the trend of linearly increasing pressure drop into a plateau was observed. Certain anomalies were observed in the trends of the trials, such as the rising pressure of the 10-inch packed sand in figure 4, where the trend between raising and lowering flowrates was very different in trend. All of these anomalies in the trends were most likely due to the packed bed sitting for too long in between flowrate increments or other such operator error.

The trend for the 10-inch packed sand in figure 4 and the 10-inch packed resin in figure 6 are observed to achieve approximately double the pressure drop of the 6-inch counterparts. It can be concluded from these trends that because of the extra height, the pressure drop will equally increase. On the contrary, the height did not have an effect on at what flowrate fluidization occurred at. The flowrate at which fluidization occurred at was different between the sand and resin, concluding that the point of fluidization matters more on the material and not height. It can also be observed from figure 6, that the 10-inch resin packed bed has the most matching trend when increasing and decreasing the flowrate.

The theory states that the raising and lowering of flowrate should not be the same. This is because of the system needing to overcome the friction coefficient when starting up. This was observed as an accurate theory for all cases except the 10-inch resin, where the speed at which flowrate was increased may have affected data. The theory also states that pressure drop will increase up until the fluidization point, and then it will reach a maximum point. Comparing the results described so far to this theory prediction, it can be concluded that the experimental trends are accurate and correct to what was expected.

References

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4. Cocco, Ray, et al. "Introduction to Fluidization." *AIChE*, Particulate Solid Research, Inc., 2014, www.aiche.org/sites/default/files/cep/20141121.pdf.

5. W. McCabe, J. Smith, and P. Harriot, *Unit Operations of Chemical Engineering*, 7th ed. McGraw-Hill, 2005.

Appendix I: Data Tabulation and Analysis

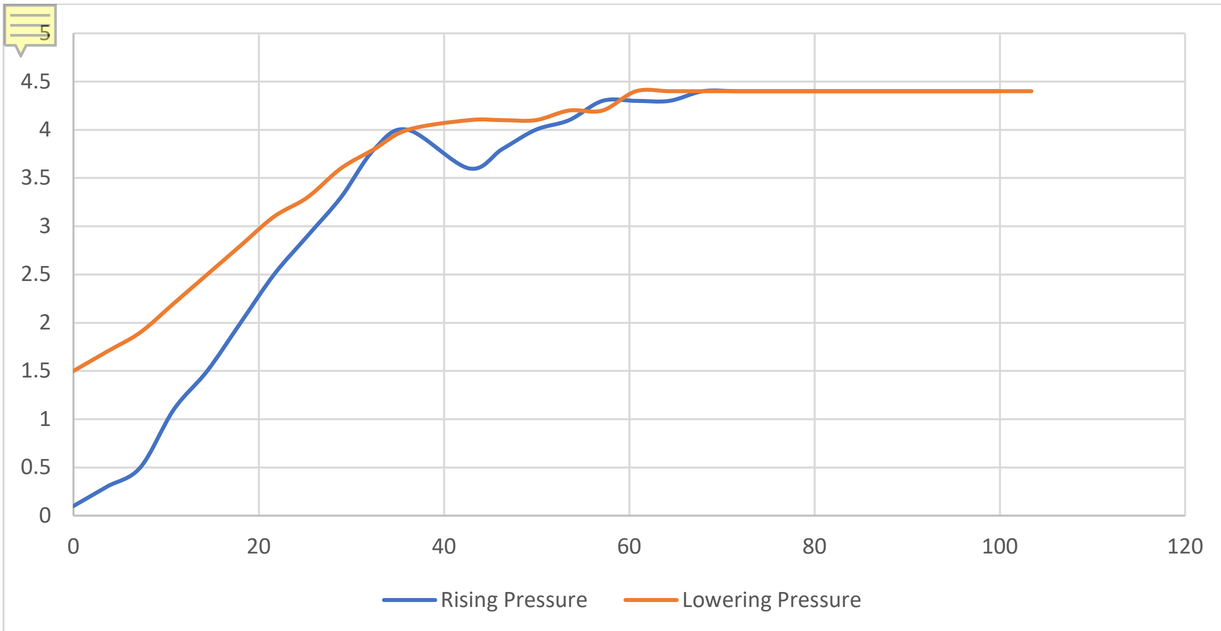

Figure 3: Pressure Drop Vs Volumetric flow rate for a 6-inch Height Packed Sand bed

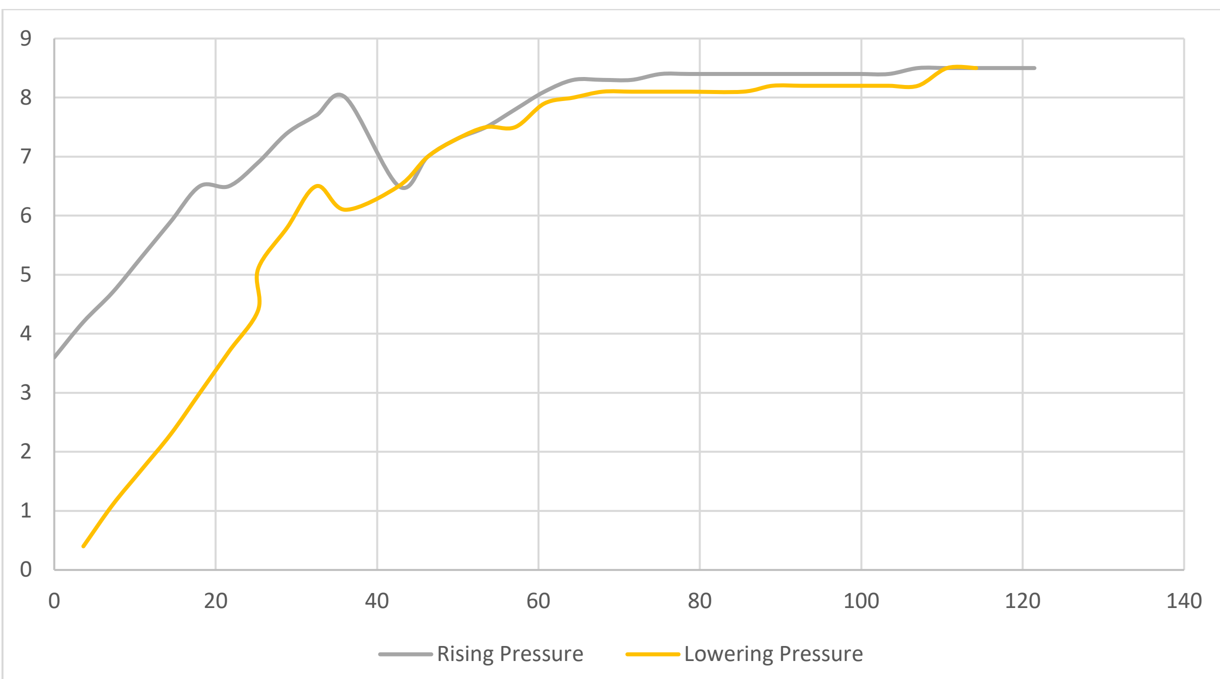

Figure 4: Pressure Drop Vs Volumetric flow rate for a 10-inch Height Packed Sand bed

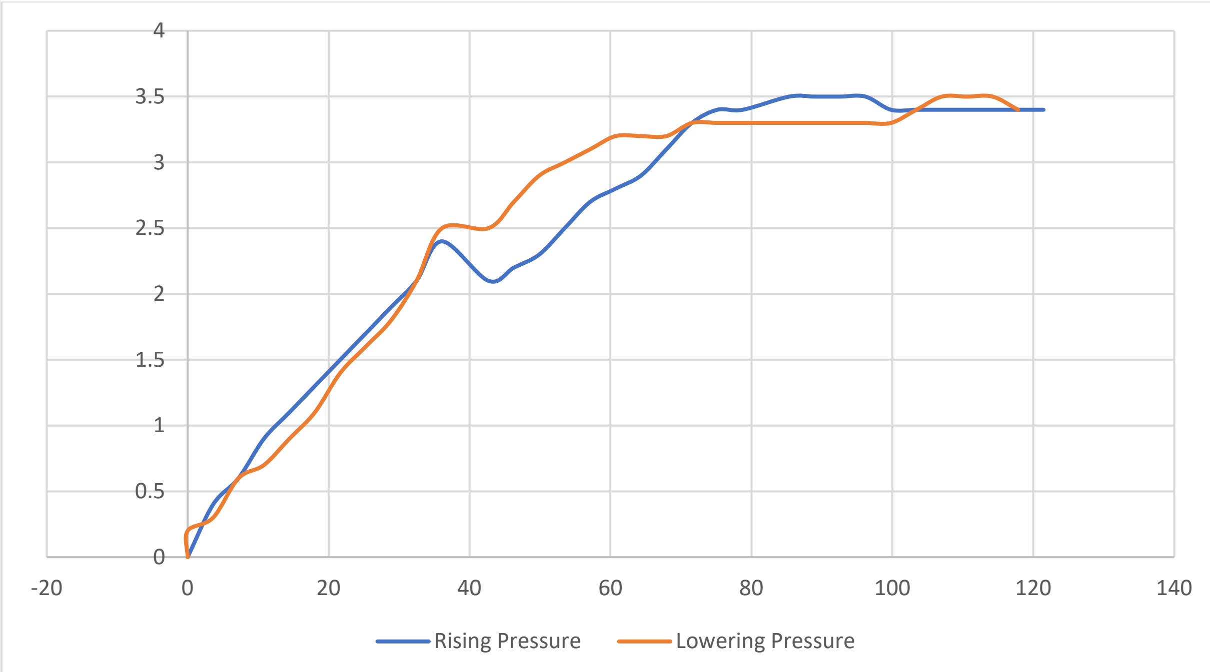

Figure 5: Pressure Drop Vs Volumetric flow rate for a 6-inch Height Packed Sand bed

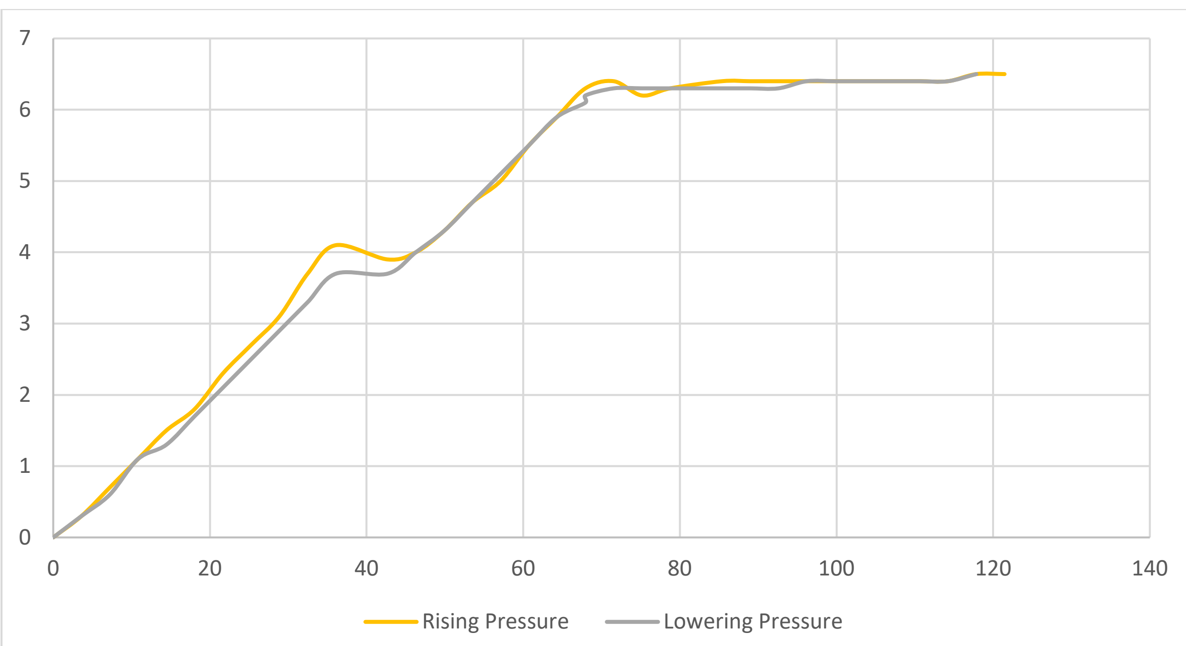

Figure 6: Pressure Drop Vs Volumetric flow rate for a 10-inch Height Packed Sand bed

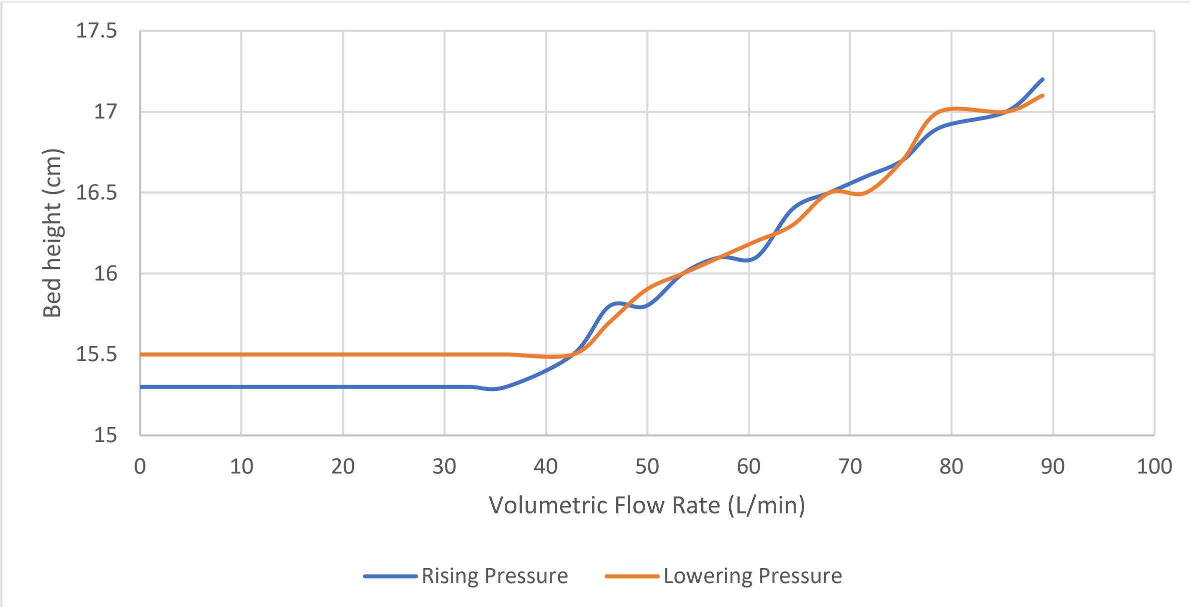

Figure 7: Bed height Vs Volumetric Flow rate for a 6 in sand bed

Table 1: Sand Pressure Table

Increasing Pressure		Decreasing Pressure	
Flow	Pressure Drop (in water)	Flow	Pressure Drop
0	0.1	0	3.6
3.609	0.3	3.609	4.2
7.218	0.5	7.218	4.7
10.827	1.1	10.827	5.3
14.436	1.5	14.436	5.9
18.045	2	18.045	6.5
21.654	2.5	21.654	6.5
25.263	2.9	25.263	6.9
28.872	3.3	28.872	7.4
32.481	3.8	32.481	7.7
36.09	4	36.09	8
42.68	3.6	42.68	6.5
46.289	3.8	46.289	7
49.898	4	49.898	7.3
53.507	4.1	53.507	7.5
57.116	4.3	57.116	7.8
60.725	4.3	60.725	8.1
64.334	4.3	64.334	8.3
67.943	4.4	67.943	8.3
71.552	4.4	71.552	8.3
75.161	4.4	75.161	8.4

78.77	4.4	78.77	8.4
85.36	4.4	85.36	8.4
88.969	4.4	88.969	8.4
92.578	4.4	92.578	8.4
96.187	4.4	96.187	8.4
99.796	4.4	99.796	8.4
103.405	4.4	103.405	8.4
99.796	4.4	107.014	8.5
96.187	4.4	110.623	8.5
92.578	4.4	114.232	8.5
88.969	4.4	117.841	8.5
85.36	4.4	121.45	8.5
78.77	4.4	117.841	8.5
75.161	4.4	114.232	8.5
71.552	4.4	110.623	8.2
67.943	4.4	107.014	8.2
64.334	4.4	103.405	8.2
60.725	4.4	99.796	8.2
57.116	4.2	96.187	8.2
53.507	4.2	92.578	8.2
49.898	4.1	88.969	8.1
46.289	4.1	85.36	8.1
42.68	4.1	78.77	8.1
36.09	4	75.161	8.1
32.481	3.8	71.552	8.1
28.872	3.6	67.943	8
25.263	3.3	64.334	7.9
21.654	3.1	60.725	7.5
18.045	2.8	57.116	7.5
14.436	2.5	53.507	7.3
10.827	2.2	49.898	7
7.218	1.9	46.289	6.5
3.609	1.7	42.68	6.1
0	1.5	36.09	6.5
		32.481	5.8
		28.872	5.1
		0	4.4
		25.263	3.7
		21.654	3
		18.045	2.3
		14.436	1.7
		10.827	1.1

Table 2: Resin Pressure Table

Rising Pressure		Lowering Pressure	
Flow	Pressure Drop	Flow	Pressure Drop
0	0	0	0
3.609	0.3	3.609	0.3
7.218	0.7	7.218	0.7
10.827	1.1	10.827	1.1
14.436	1.5	14.436	1.5
18.045	1.8	18.045	1.8
21.654	2.3	21.654	2.3
25.263	2.7	25.263	2.7
28.872	3.1	28.872	3.1
32.481	3.7	32.481	3.7
36.09	4.1	36.09	4.1
42.68	3.9	42.68	3.9
46.289	4	46.289	4
49.898	4.3	49.898	4.3
53.507	4.7	53.507	4.7
57.116	5	57.116	5
60.725	5.5	60.725	5.5
64.334	5.9	64.334	5.9
67.943	6.3	67.943	6.3
71.552	6.4	71.552	6.4
75.161	6.2	75.161	6.2
78.77	6.3	78.77	6.3
85.36	6.4	85.36	6.4
88.969	6.4	88.969	6.4
92.578	6.4	92.578	6.4
96.187	6.4	96.187	6.4
99.796	6.4	99.796	6.4
103.405	6.4	103.405	6.4
107.014	6.4	107.014	6.4
110.623	6.4	110.623	6.4
114.232	6.4	114.232	6.4
117.841	6.5	117.841	6.5
121.45	6.5	121.45	6.5
117.841	6.5	117.841	6.5
114.232	6.4	114.232	6.4
110.623	6.4	110.623	6.4
107.014	6.4	107.014	6.4
103.405	6.4	103.405	6.4

99.796	6.4	99.796	6.4
96.187	6.4	96.187	6.4
92.578	6.3	92.578	6.3
88.969	6.3	88.969	6.3
85.36	6.3	85.36	6.3
78.77	6.3	78.77	6.3
75.161	6.3	75.161	6.3
71.552	6.3	71.552	6.3
67.943		0	
64.334	6.1	67.943	6.1
60.725	5.9	64.334	5.9
57.116	5.5	60.725	5.5
53.507	5.1	57.116	5.1
49.898	4.7	53.507	4.7
46.289	4.3	49.898	4.3
42.68	4	46.289	4
36.09	3.7	42.68	3.7
32.481	3.7	36.09	3.7
28.872	3.3	32.481	3.3
25.263	2.9	28.872	2.9
21.654	2.5	25.263	2.5
18.045	2.1	21.654	2.1
14.436	1.7	18.045	1.7
10.827	1.3	14.436	1.3
7.218	1.1	10.827	1.1
3.609	0.6	7.218	0.6

Appendix II: Error Analysis

Fluidized Bed Apparatus

There were several sources of error as a result of limitations encountered over the course of the experiment:

1. **Inconsistent gas flow rate:** As the equipment had manual gauges instead of digital gauges, there was a need to ensure that the flow rate adjustment was consistent by setting the middle of the carrier ball within the meter to the desired marking. However, this was not easily achieved as the valve used to control the inlet gas flow rate was extremely sensitive.
2. **House Gas Interruption:** In the middle of the resin bed experiment, the house gas supply was disrupted, leading to insufficient gas was entering the apparatus. A temporary halt had to be observed while technicians rectified the issue. As a result, pressure drop readings were slightly compromised.
3. **Bed Level Inconsistencies:** While filling the column with the packing, it was found that the packing was not able to be distributed evenly across the diameter of the column. Efforts were made to knock the sand/resin into place to ensure that they could be as well packed as possible. Therefore, the 6 and 10 inches as outlined in the experimental design were in reality, less than the 6 and inches expected respectively. However, this inconsistency was negligible since the variable for this experiment had to do with the inlet gas velocity.

The three discussed points, however, did have an impact on the superficial velocity. In the Ergun equation, the superficial velocity (theoretical) for a sand packed bed was calculated to be 0.122 m/s and the actual measured superficial velocity was found to be 0.119 m/s.

$$\%\text{Error} = \frac{\text{actual} - \text{theoretical}}{\text{theoretical}} \times 100\%$$

Using the equation outlined above, the %error calculated for the experimental data are listed in table [3]

Table 3: % Error calculated for various column configurations

Bed Height	**Sand**	**Resin**
6 in	2.46%	59.4%
10 in	9.84%	53.9%

From the results, the sensitivity of the apparatus was more impactful on the resin bed as compared to the sand bed. It was also important to mention that the column was packed with the resin when the supply of the house gas was disrupted. As a result, a high %error for the resin bed was expected. This was not the case for the sand packed bed, since errors were calculated to within a tolerance level of 10%.

Void Fraction Determination

There were only a few possible sources of error when determining the void fraction. Experimentally determining the void fraction is a fairly inaccurate method. The possible errors include:

1. **The water took too long to flow through all of the particles:** once the water was mixed with the particles, it seemed to flow only halfway through the particles and then stall. The water eventually reached to the bottom of the particles after a few more minutes.

Appendix III: Sample Calculations

Average Particle Size:

$$D_{pavg} = \frac{1}{\sum \frac{x_i}{D_{pi}}}$$

x_i = mass fraction
D_{pi} = Particle Size (This was found by using sifters. The smallest pore sifter that still allowed sand and resin beads to pass through were recorded)

mL	Total mass	Sand Density kg/m^3
320	0.51264	1602

Volume (mL)	mass (kg)	Dpi (m)	xi	xi/Dpi	DpAvg
100	0.1602	0.0005	0.3125	625	0.0005
50	0.0801	0.0005	0.15625	312.5	
25	0.04005	0.0005	0.078125	156.25	
5	0.00801	0.0005	0.015625	31.25	
40	0.06408	0.0005	0.125	250	
100	0.1602	0.0005	0.3125	625	

The average sand particle size was calculated to be

D_{pavg}= 500 $\mu m/kg$

mL	Total mass	Resin Bead Density kg/m^3
275	0.264	960

Volume	mass	Dpi	xi	xi/Dpi	DpAvg
100	0.096	0.000841	0.36363636	432.385688	0.000841
50	0.048	0.000841	0.18181818	216.192844	
25	0.024	0.000841	0.09090909	108.096422	
100	0.096	0.000841	0.36363636	432.385688	

The average sand particle size was calculated to be

D_{pavg}= 841 $\mu m/kg$

Void Fraction:

$$\varepsilon_{m(S)} = \frac{\text{Volume of Void Space}}{\text{Total Volume}} = \frac{60}{160} = .375$$

$$\varepsilon_{m(R)} = \frac{\text{Volume of Void Space}}{\text{Total Volume}} = \frac{80}{180} = .444$$

Minimum superficial Velocity for Sand

$$V_o \approx \frac{g(\rho_p - \rho)}{150\mu}\frac{\varepsilon_m^3}{1-\varepsilon_m}\Phi_s^2 D_p^2 = \frac{9.81\frac{m}{s^2}\left(1602\frac{kg}{m^3} - 1.225\frac{kg}{m^3}\right)\frac{(.375)^3}{1-.375}}{150\left(.0000181\frac{kg}{m*s}\right)}(1)^2(.0005\ m)^2 = .122\frac{m}{s}$$

Minimum superficial Velocity for Resin

$$V_o \approx \frac{g(\rho_p - \rho)}{150\mu}\frac{\varepsilon_m^3}{1-\varepsilon_m}\Phi_s^2 D_p^2 = \frac{9.81\frac{m}{s^2}\left(960\frac{kg}{m^3} - 1.225\frac{kg}{m^3}\right)\frac{(.444)^3}{1-.444}}{150\left(.0000181\frac{kg}{m*s}\right)}(1)^2(.000841\ m)^2 = .386\frac{m}{s}$$

The minimum velocity for a 6 inch filled bed with sand

$$57.116\frac{L}{min} * \frac{1\ min}{60\ sec} * \frac{1\ m^3}{1000\ L} * \frac{1}{.008\ m^3} = .119\frac{m}{s}$$

The minimum velocity for a 10 inch filled bed with sand

$$64.33\frac{L}{min} * \frac{1\ min}{60\ sec} * \frac{1\ m^3}{1000\ L} * \frac{1}{.008\ m^3} = .1340\frac{m}{s}$$

The minimum velocity for a 6 inch filled bed with Resin

$$75.161\ \frac{L}{min} * \frac{1\ min}{60\ sec} * \frac{1\ m^3}{1000\ L} * \frac{1}{.008\ m^3} = .1566\frac{m}{s}$$

The minimum velocity for a 10 inch filled bed with Resin

$$85.36\ \frac{L}{min} * \frac{1\ min}{60\ sec} * \frac{1\ m^3}{1000\ L} * \frac{1}{.008\ m^3} = .1778\frac{m}{s}$$

Appendix IV: Individual Contributions

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6.15	All
Pre-Lab Editing	1	All
Experimental Objective (Prelab)	0.5	Allison
Apparatus (prelab)	0.25	Allison
Materials & Supplies (prelab)	0.5	Ian
Procedure (prelab)	0.75	Peter
Project Safety Evaluation (prelab)	0.5	Emily
Final Lab Editing	1.5	All
Executive Summary (final lab)	1	Emily
Introduction (Final lab)	1	Emily
Theory (Final Lab)	3	Ian
Results (final lab)	6.5	Allison
Discussion (final lab)	5	Peter
References (final and/or prelab)	0.5	All
Data Tabulation/Graphs (final)	2	Allison
Error Analysis (final lab)	2	Ian
Sample Calculations (final Lab)	3	Allison
Power Point Presentation	2.5	All
Total	37.65	-